CyberSecurity Lab1

Part 1 : understanding Password Security

Step1 : Analyse Weak Passwords

Discussion

1. Why are these passwords insecure?

   Passwords such as 123456, password, admin, 12345678, and qwerty are insecure because they are very common and easy to guess. They are short and do not contain a strong mix of uppercase letters, lowercase letters, numbers, and special characters. Because many people use them, attackers can easily guess them using automated tools.

2. How quickly could a computer guess these passwords?

   A computer can guess these passwords very quickly, sometimes in a few seconds. Attackers use automated tools that test thousands or even millions of password combinations in a short time. Since these passwords are common, they are usually cracked almost immediately.

3. Attackers often use automated tools to try many passwords quickly. A dictionary attack tests a list of common passwords, while a brute force attack tries many possible combinations until the correct password is found. These methods make weak passwords easy to crack.

Step 2: Create Strong Passwords

Let's write my examples below:

1. Cyb3r$Shield_92
2. Str0ng!P@ssKey45
3. Gil$ecur3_Lab88

Part 2: Observing Network Traffic

Step 1: Launch Wireshark

Step 2: Generate Network Traffic

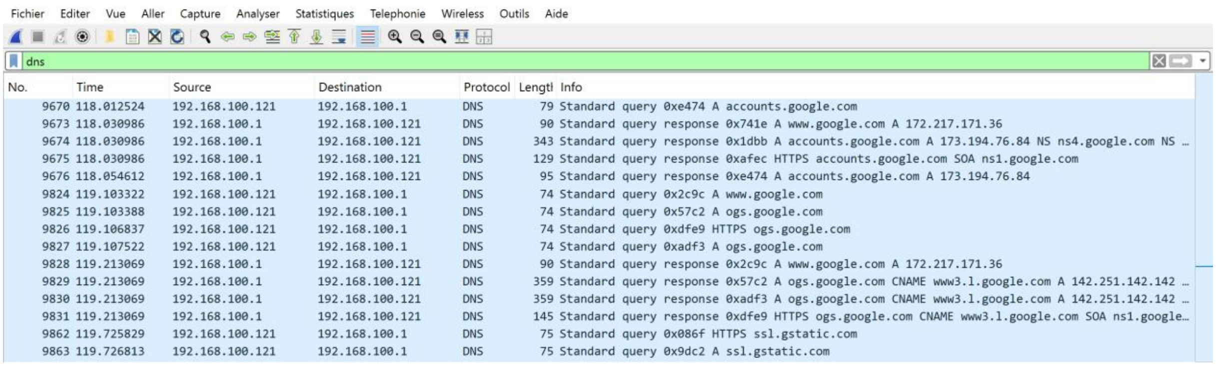
Step 3: Stop Packet Capture

Part 3: Packet Analysis

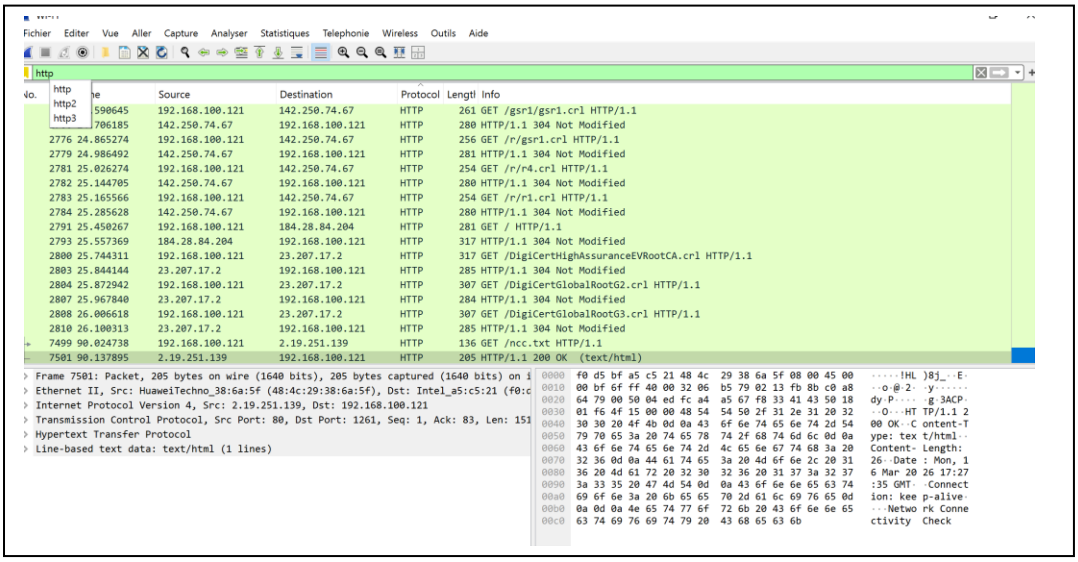
Step 1: DNS Traffic

Let's write one domain name observed:

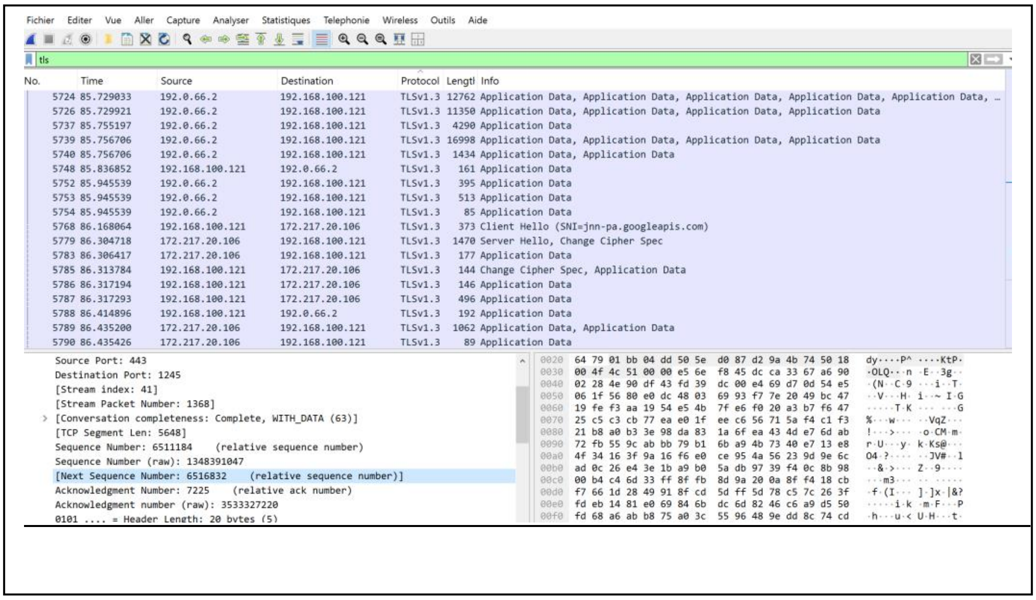
<u>www.google.com</u>



<u>Step 2: HTTP Traffi</u>



Step 3: Encrypted Traffic



Answers to the questions

1. Approximately 1500 packets were captured during the experiment.
2. Let's list three protocols observed during the experiment:
   DNS, HTTP, TLS
3. Encryption is important when browsing the internet because it protects sensitive information transmitted over the Internet. Without encryption, data such as passwords, personal information, or financial data could be intercepted and read by attackers.

4. Let's show the types of information could attackers obtain from unencrypted traffic:
   An attacker could access several types of sensitive information, including login credentials, personal data, session cookies, email addresses, requests sent to websites, or even the pages visited by the user.

Lab Report

This experiment clearly shows that network traffic can be easily captured and analyzed with accessible tools. It also demonstrates that weak passwords represent a major vulnerability and that unencrypted communications can expose sensitive information. Therefore, the use of encryption and strong passwords is essential to protect users' security and privacy on the Internet.